

Digital Assignment - ①

Object oriented Program :-

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① Explain the following Terms :-

Ans

Object :

① Objects are the basic run time entities in an object-oriented system.

② They may represent a person, place, a bank account, a table of data or any item that a program has to handle.

③ Program objects should be chosen such that they match closely with the real-world objects.

Example :- Table, fan, vehicle, pen, keyboard etc..

Classes :-

① Object contain data, and code to manipulate that data. The entire set of data and code of an object can be made a user-defined datatype with the help of class.

② A class is the collection of objects.

- ④ Each object is associated with data and type class with which they are created -

Example :- Mango, Apple and Orange are the members of class fruit.

Data Encapsulation :-

- ④ Wrapping up of data and function into a single unit is known as Encapsulation.
- ④ Data and Encapsulation is the most striking features of a class.
- ④ The data is not accessible to the outside world, and only those function which are wrapped in a class are accessible.

Example :- object : STUDENT

DATA :

Name, dob, marks.

functions :

total, Avg, display.

Information hiding :-

- ④ Information hiding is a fundamental principle that emphasizes encapsulation to control the access to the internal details of an object.

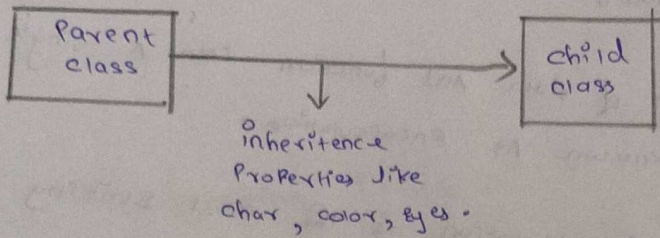
Examples of information hiding are,

Encapsulation, data modifiers, Abstraction.

Inheritance :-

- ① inheritance is the process by which objects of one class is acquired the properties of objects of another class.

for example :-



- ② the concept of inheritance is "Reusability".

Polyorphism :-

- ① the ability to make more than one form. A operation may exhibit different behaviour in different instances.
- ② using a single function name to perform different types of task is known as function overloading.

for ex :

for two numbers, the operation will generate a sum,
if the operands are strings, then the operation would produce a third string by concatenation.

C++ Program to find sum of series using do-while
 $1^2 + 3^2 + 5^2 + \dots + n^2$.

Program :-

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int number, result = 0, count = 1;
```

```
    cout << "Enter A Number :";
```

```
    cin >> number;
```

```
    do {
```

```
        result += count * count
```

```
        count += 2;
```

```
    } while (count <= number);
```

```
    cout << "Sum = " << result;
```

```
    return 0;
```

```
}
```

Output :-

Input : 5

Output : 35

2) Write a C++ Program to read a number n and digit d and check whether d is present in the number n . If it is find out the position of d in number n .

Program :-

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int number, len_str;
```

```
    cout << "Enter A Number :";
```



```

cin >> number ;

string str_number = to_string (number) ;

len_str = str_number.length () ;

char read_num ;

cout << "Enter A number to check wheather it is Present : ";

cin >> read_num ;

for ( int i=0 ; i< len_str ; i++) {

    if ( str_number[i] == read_num ) {

        cout << "read_num << " is present At the position " << i
        << " from left to right " ;

        break ;

    }

    else {

        cout << read_num << " is not Present " ;

        break ;

    }

}

return 0 ;
}

```

Output :-

input n : 10235

Output : 5 is Present at the position 4 from left to right

d : 5.

④ Summarize the syntactic rules of the following loop statements

① for loop.

② while loop.

③ do while.

for loop :-

- ① for loop is A most comfortable loop.
- ② it executes A specified block of code for A fixed Number of iterations, with explicit Initialization, Condition, And update expressions.

Syntax :-

```
for (Initialization; condition; update) {  
    // Block of code  
}
```

Example :-

```
for (int i = 0; i < 5; i++) {  
    cout << i << " ";  
}
```

while loop :-

- ① Repeatedly executes A block of code as long as the specified conditions checked before each iteration.

Syntax :-

```
while (condition) {  
    // code to be executed &
```

Example :-

```
counter = 1  
while (counter <= 5) {  
    cout << counter;  
    counter ++
```

```
}
```

do-while - loop :-

⊛ It is similar to a while loop but guarantees at least one execution of the code blocks since the condition is checked after the first iteration.

Syntax :-

do {

// code block

} while (condition);

Example :-

int counter = 1;

do {

cout << counter << "endl";

counter ++;

} while (counter <= 5);